Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Function transformations**

Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

$g(0) = -3(2)^2 - 4 = -3(4) - 4 = -16$. g is f shifted left 2, reflected over the x-axis and stretched vertically by a factor of 3, then shifted down 4.

2 **[Foundational]** • **Target: Function transformations**

Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g's graph.

$g(-2) = -3(0)^2 - 4 = -4$. Since $f(x) = x^2$ has its vertex (minimum) at $x = 0$, and g shifts f left by 2, the transformed vertex sits at $x = -2$.

3 **[Foundational]** • **Target: Even and odd functions**

Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

Only even-degree terms appear (x^4, x^2, and the constant), so by Theorem 3.1 f is even. Check: $f(2) = 16 + 8 - 5 = 19$ and $f(-2) = 16 + 8 - 5 = 19$ ✓.

4 **[Foundational]** • **Target: Even and odd functions**

Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.

Only odd-degree terms appear (x^5, x), so by Theorem 3.1 h is odd. Check: $h(1) = 2 - 3 = -1$ and $h(-1) = -2 + 3 = 1 = -h(1)$ ✓.

5 **[Foundational]** • **Target: Even and odd functions**

Explain why $k(x) = x^2 + x$ is neither even nor odd.

k mixes an even-degree term (x^2) with an odd-degree term (x), so Theorem 3.1 doesn't apply to the whole function. Checking directly: $k(1) = 2$ and $k(-1) = 0$, which is neither equal to $k(1)$ (even) nor $-k(1) = -2$ (odd).

6 **[Foundational]** • **Target: Domain restrictions**

Find the domain of $f(x) = \sqrt{2x + 6}$.

Need $2x + 6 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

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7 [Foundational] • Target: Domain restrictions

Find the domain of $g(x) = \frac{1}{x^2 - 25}$.

Need $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$. Domain: all reals except 5 and -5.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise functions

Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.

Since $-2 < 1$, use the first piece: $p(-2) = 2(-2) + 3 = -1$. Since $3 \geq 1$, use the second piece: $p(3) = 3^2 - 1 = 8$.

9 [Intermediate] • Target: Function transformations

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. We need to solve the quadratic equation $x^2 - 4x + 3 = 0$. This equation can be factored as $(x - 1)(x - 3) = 0$. Therefore, the solutions are $x = 1$ and $x = 3$. This means the x-intercepts are $(1, 0)$ and $(3, 0)$.

10 [Intermediate] • Target: Even and odd functions

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?

The y-intercept of a function's graph is the point where the graph crosses the y-axis. This occurs when $x = 0$. To find the y-intercept, substitute $x = 0$ into the function: $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the y-intercept is 3.

11 [Intermediate] • Target: Domain restrictions

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?

The value of $f(0)$ is the y-coordinate of the point where the graph of the function intersects the y-axis. From the graph, we can see that the function intersects the y-axis at the point $(0, -1)$. Therefore, $f(0) = -1$.

12 [Intermediate] • Target: Piecewise functions

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

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13 **[Intermediate]** • **Target: Rational Functions & Oblique Asymptotes**

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

The function is $f(x) = x^3 - 6x^2 + 5$. The derivative is $f'(x) = 3x^2 - 12x$. At $x = a = 1$, the slope of the tangent line is $m = f'(1) = 3(1)^2 - 12(1) = 3 - 12 = -9$. The y-coordinate of point P is $f(1) = 1^3 - 6(1)^2 + 5 = 1 - 6 + 5 = 0$. So, $P = (1, 0)$. The equation of the tangent line at P is $y - 0 = -9(x - 1)$, which simplifies to $y = -9x + 9$. To find the intersection points of the tangent line and the curve, we set the function equal to the tangent line equation: $x^3 - 6x^2 + 5 = -9x + 9$. Rearranging this equation gives $x^3 - 6x^2 + 9x - 4 = 0$. We know that $x = 1$ is a root because it is the point of tangency, meaning it is a repeated root (at least a double root). Therefore, $(x - 1)^2$ must be a factor of the polynomial. Let's perform polynomial division or synthetic division. Since $x = 1$ is a root, $(x - 1)$ is a factor. $x^3 - 6x^2 + 9x - 4 = (x - 1)(x^2 - 5x + 4)$. Factoring the quadratic term, we get $x^2 - 5x + 4 = (x - 1)(x - 4)$. Thus, the equation becomes $(x - 1)(x - 1)(x - 4) = 0$, or $(x - 1)^2(x - 4) = 0$. The roots are $x = 1$ (a double root) and $x = 4$. The point Q is the other intersection point, so its x-coordinate is $b = 4$.

14 **[Intermediate]** • **Target: Piecewise & Absolute Value Graphs**

Find the domain of $f(x) = \sqrt{x - 9} + \frac{1}{x - 11}$.

The square root needs $x \geq 9$; the fraction needs $x \neq 11$ — both together (and $9 \neq 11$ so neither condition swallows the other). Final Answer: $x \geq 9,\ x \neq 11$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Inverse Trigonometric Graphs & Domains • Cross-Lesson: 2.01 Sequences & Series

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

Let X be the number of defective bulbs in a batch of $n = 5$. We are given that X follows a binomial distribution with parameters $n = 5$ and probability of success (a bulb being defective) p. The probability of exactly k defective bulbs is $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. We are given that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Therefore, $P(X = 2) = \binom{5}{2} p^2 (1-p)^{5-2} = \frac{10}{32}$. First, calculate the binomial coefficient: $\binom{5}{2} = \frac{5!}{2!\,(5-2)!} = \frac{5!}{2!\,3!} = \frac{5 \times 4}{2 \times 1} = 10$. So, the equation becomes: $10p^2(1-p)^3 = \frac{10}{32}$. Divide both sides by 10: $p^2(1-p)^3 = \frac{1}{32}$. We need to find a value of p that satisfies this equation. Let's test the given options: If $p = 0.2$: $(0.2)^2(1-0.2)^3 = (0.04)(0.8)^3 = 0.04 \times 0.512 = 0.02048$. $\frac{1}{32} = 0.03125$. This is not correct. If $p = 0.5$: $(0.5)^2(1-0.5)^3 = (0.25)(0.5)^3 = 0.25 \times 0.125 = 0.03125$. $\frac{1}{32} = 0.03125$. This is correct. If $p = 0.8$: $(0.8)^2(1-0.8)^3 = (0.64)(0.2)^3 = 0.64 \times 0.008 = 0.00512$. This is not correct. If $p = 0.75$: $(0.75)^2(1-0.75)^3 = (0.5625)(0.25)^3 = 0.5625 \times 0.015625 = 0.0087890625$. This is not correct. Thus, the value of p is 0.5.

16 [Challenging] • Target: Transformations of Non-Linear Functions • Cross-Lesson: 4.01 Matrices & Vectors

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?

The transformation grid shows a reflection across the line $y = x$. When a point (x, y) is reflected across the line $y = x$, its image has coordinates (y, x). Given point A is $(1, 2)$, its image A' will have coordinates $(2, 1)$. However, the provided `transformation_grid` JSON shows an object at `[[1, 2], [4, 2], [2, 5]]` labeled A, B, C and its image at `[[-4, 2], [-1, 2], [-3, 5]]` labeled A', B', C'. This implies a translation, not a reflection. Let's analyze the transformation of point A. Original point A is at $(1, 2)$. Its image A' is at $(-4, 2)$. To find the translation vector $(\Delta x\ \Delta y)$, we subtract the original coordinates from the image coordinates: $\Delta x = -4 - 1 = -5$ and $\Delta y = 2 - 2 = 0$. So the translation is by the vector $(-5\ 0)$. The question asks for the coordinates of A' given A. Based on the figure's data, A' is $(-4, 2)$. The provided choices are: (-4, 2), (-2, 1), (2, -4), (1, -2). The first choice matches the figure.

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17 [Challenging] • Target: Function transformations • Cross-Lesson: 5.01 Introductory Calculus

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

Let X_A be the number of defective components of type A, and X_B be the number of defective components of type B. The total number of defective components is $X = X_A + X_B$. The expected number of defective components of type A is $E[X_A] = N \times 0.03$. The expected number of defective components of type B is $E[X_B] = (100 - N) \times 0.05$. The total expected number of defective components is $E[X] = E[X_A] + E[X_B]$. We are given $E[X] = 4.2$. So, $N \times 0.03 + (100 - N) \times 0.05 = 4.2$. $0.03N + 5 - 0.05N = 4.2$. $5 - 0.02N = 4.2$. $0.02N = 5 - 4.2$. $0.02N = 0.8$. $N = \frac{0.8}{0.02} = 40$. Therefore, there are 40 components of type A and $100 - 40 = 60$ components of type B. The question asks for the probability that a randomly selected component from this batch is defective, given that it is of type A. This is simply the defect rate of type A components. Probability (Defective | Type A) = Defect rate of Type A = 0.03. Final Answer: 0.030

18 [Challenging] • Target: Even and odd functions • Cross-Lesson: 6.01 Probability

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is 6. We are interested in the event of rolling a number greater than 4. The numbers in the set of outcomes that are greater than 4 are {5, 6}. There are 2 favorable outcomes. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Probability (rolling > 4) = (Number of favorable outcomes) / (Total number of possible outcomes) = 2 / 6 = 1/3.

19 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

A matrix $M = (k, 2; 3, 1)$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of k.

The transformation of a point $P(x, y)$ by a matrix $M = (a, b; c, d)$ is given by MP. In this case, we have $(k, 2; 3, 1)(1; 2) = (5; 5)$. Performing the matrix multiplication, we get: $k(1) + 2(2) = 5$ and $3(1) + 1(2) = 5$. The second equation, $3 + 2 = 5$, is consistent. For the first equation, we have $k + 4 = 5$. Solving for k, we subtract 4 from both sides: $k = 5 - 4$, which gives $k = 1$. Therefore, the value of k is 1.

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20 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

The total number of marbles in the bag is the sum of red and blue marbles: $5 + 3 = 8$. The number of favorable outcomes (drawing a blue marble) is 3. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a blue marble is $\frac{\text{Number of blue marbles}}{\text{Total number of marbles}} = \frac{3}{8}$. Final Answer: $\frac{3}{8}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function transformations

Classify $f(x) = 2x^3 + 5$ as even, odd, or neither.

a. even

b. both even and odd

c. **neither ✓**

d. odd

$f(-x) = 2(-x)^3 + 5$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

22 [Foundational] • Target: Even and odd functions

Classify $f(x) = -3x^5 + 6x$ as even, odd, or neither.

a. both even and odd

b. even

c. neither

d. **odd ✓**

$f(-x) = -3(-x)^5 + 6(-x) = -f(x)$, so f is odd.

23 [Foundational] • Target: Domain restrictions

Classify $f(x) = -x^2 + 6$ as even, odd, or neither.

a. both even and odd

b. neither

c. odd

d. **even ✓**

$f(-x) = -(-x)^2 + 6 = f(x)$, so f is even.

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24 [Foundational] • Target: Piecewise functions

Classify $f(x) = 3x^5 + 6x$ as even, odd, or neither.

a. even

b. neither

c. **odd** ✓

d. both even and odd

$f(-x) = 3(-x)^5 + 6(-x) = -f(x)$, so f is odd.

25 [Foundational] • Target: Rational Functions & Oblique Asymptotes

Classify $f(x) = -2x^3 + 8$ as even, odd, or neither.

a. **neither** ✓

b. odd

c. both even and odd

d. even

$f(-x) = -2(-x)^3 + 8$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

26 [Foundational] • Target: Piecewise & Absolute Value Graphs

Classify $f(x) = -3x^3 + 4x$ as even, odd, or neither.

a. even

b. neither

c. both even and odd

d. **odd** ✓

$f(-x) = -3(-x)^3 + 4(-x) = -f(x)$, so f is odd.

27 [Foundational] • Target: Inverse Trigonometric Graphs & Domains

Classify $f(x) = -x^4 + 8x^2$ as even, odd, or neither.

a. odd

b. both even and odd

c. **even** ✓

d. neither

$f(-x) = -(-x)^4 + 8(-x)^2 = f(x)$, so f is even.

28 [Foundational] • Target: Transformations of Non-Linear Functions

Find the domain of $f(x) = \sqrt{5x - 35}$.

a. $x \geq 9$

b. $x \geq 7$ ✓

c. $x \geq 5$

d. $x \leq 7$

Need $5x - 35 \geq 0 \Rightarrow x \geq 7$. Domain: $x \geq 7$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Function transformations**

Find the domain of $f(x) = \sqrt{4x - 24}$.

a. $x \geq 4$

b. $x \leq 6$

c. $x \geq 8$

d. $x \geq 6$ ✓

Need $4x - 24 \geq 0 \Rightarrow x \geq 6$. Domain: $x \geq 6$.

30 **[Intermediate]** • **Target: Even and odd functions**

Find the domain of $f(x) = \sqrt{x + 3}$.

a. $x \leq -3$

b. $x \geq -5$

c. $x \geq -3$ ✓

d. $x \geq -1$

Need $x + 3 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

31 **[Intermediate]** • **Target: Domain restrictions**

Find the domain of $f(x) = \sqrt{5x + 25}$.

a. $x \geq -5$ ✓

b. $x \geq -7$

c. $x \leq -5$

d. $x \geq -3$

Need $5x + 25 \geq 0 \Rightarrow x \geq -5$. Domain: $x \geq -5$.

32 **[Intermediate]** • **Target: Piecewise functions**

Find the domain of $f(x) = \sqrt{5x}$.

a. $x \geq 0$ ✓

b. $x \geq 2$

c. $x \leq 0$

d. $x \geq -2$

Need $5x \geq 0 \Rightarrow x \geq 0$. Domain: $x \geq 0$.

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33 [Intermediate] • Target: Rational Functions & Oblique Asymptotes

Find the domain of $f(x) = \sqrt{-2x + 12}$.

a. $x \leq 6$ ✓

b. $x \leq 8$

c. $x \leq 4$

d. $x \geq 6$

Need $-2x + 12 \leq 0 \Rightarrow x \leq 6$. Domain: $x \leq 6$.

34 [Intermediate] • Target: Piecewise & Absolute Value Graphs

Find the domain of $f(x) = \sqrt{-3x - 27}$.

a. $x \leq -11$

b. $x \geq -9$

c. $x \leq -7$

d. $x \leq -9$ ✓

Need $-3x - 27 \leq 0 \Rightarrow x \leq -9$. Domain: $x \leq -9$.

35 [Intermediate] • Target: Inverse Trigonometric Graphs & Domains

Let $p(x) = 3x - 6$ for $x < 3$, and $p(x) = -2x + 3$ for $x \geq 3$. Find $p(-2)$ and $p(7)$.

a. $-12, -11$ ✓

b. $-12, -13$

c. $-10, -11$

d. $-11, -12$

Since $-2 < 3$, use the first piece: $p(-2) = 3(-2) - 6 = -12$. Since $7 \geq 3$, use the second piece: $p(7) = -2(7) + 3 = -11$.

36 [Intermediate] • Target: Transformations of Non-Linear Functions

Let $p(x) = 4x - 7$ for $x < 5$, and $p(x) = -x + 1$ for $x \geq 5$. Find $p(3)$ and $p(5)$.

a. $5, -6$

b. $5, -4$ ✓

c. $-4, 5$

d. $7, -4$

Since $3 < 5$, use the first piece: $p(3) = 4(3) - 7 = 5$. Since $5 \geq 5$, use the second piece: $p(5) = -(5) + 1 = -4$.

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37 [Intermediate] • Target: Function transformations

Let $f(x) = x^2$ and $g(x) = 4f(x-3) - 1 = 4(x-3)^2 - 1$. Find $g(4)$.

a. 7

b. -1

c. -3

d. 3 ✓

$g(4) = 4(1)^2 - 1 = 4(1) - 1 = 3$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Even and odd functions • Cross-Lesson: 6.01 Probability

Let $f(x) = x^2$ and $g(x) = -f(x-2) + 8 = -(x-2)^2 + 8$. Find $g(-8)$.

a. -96

b. -92 ✓

c. -88

d. 92

$g(-8) = -(-10)^2 + 8 = -(100) + 8 = -92$.

39 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

Find the domain of $g(x) = \frac{1}{x^2 - 144}$.

a. $14, -12$

b. $12, -12$ ✓

c. $-12, 12$

d. $12, -14$

Need $x^2 - 144 \neq 0 \Rightarrow x \neq \pm 12$. Domain: all reals except 12 and -12.

40 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

Find the domain of $g(x) = \frac{1}{x^2 - 49}$.

a. $9, -7$

b. $7, -9$

c. $7, -7$ ✓

d. $-7, 7$

Need $x^2 - 49 \neq 0 \Rightarrow x \neq \pm 7$. Domain: all reals except 7 and -7.

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41 [Challenging] • Target: Rational Functions & Oblique Asymptotes • Cross-Lesson: 4.01 Matrices & Vectors

Find the oblique asymptote of $f(x) = \frac{3x^2 - 2x + 4}{x - 1}$.

a. $y = 3x + 1$ ✓

b. $y = 3x - 1$

c. $y = 3x + 4$

d. $y = 3x$

Divide: $\frac{3x^2 - 2x + 4}{x - 1} = 3x + 1 + \frac{5}{x - 1}$. The oblique asymptote is $y = 3x + 1$.

42 [Challenging] • Target: Rational Functions & Oblique Asymptotes • Cross-Lesson: 5.01 Introductory Calculus

What is the horizontal asymptote of $h(x) = \frac{5x - 2}{x^2 + 1}$?

a. $y = 0$ ✓

b. $y = 5$

c. $y = -2$

d. No horizontal asymptote

Since the degree of the denominator (2) is strictly greater than the degree of the numerator (1), the horizontal asymptote is $y = 0$.

43 [Challenging] • Target: Piecewise & Absolute Value Graphs • Cross-Lesson: 6.01 Probability

What is the domain of $f(x) = \sqrt{9 - x^2}$?

a. $[-3, 3]$ ✓

b. $(-\infty, -3] \cup [3, \infty)$

c. $[0, 3]$

d. $[-9, 9]$

Requirement: $9 - x^2 \geq 0 \Rightarrow x^2 \leq 9 \Rightarrow -3 \leq x \leq 3$.

44 [Challenging] • Target: Inverse Trigonometric Graphs & Domains • Cross-Lesson: 1.01 Complex Numbers

What is the range of the principal $\arccos(x)$ function?

a. $[0, \pi]$ ✓

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, 2\pi]$

d. $(0, \pi)$

By standard mathematical convention, the principal branch of $\arccos(x)$ has range $[0, \pi]$.

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45 [Challenging] • Target: Transformations of Non-Linear Functions • Cross-Lesson: 2.01 Sequences & Series

How is the graph of $y = 2f(-x) + 3$ obtained from $y = f(x)$?

a. Reflect over y-axis, vertical stretch by 2, shift up 3 ✓

b. Reflect over x-axis, vertical stretch by 2, shift up 3

c. Reflect over y-axis, horizontal stretch by 2, shift up 3

d. Reflect over origin, shift up 3

$f(-x)$ reflects across the y-axis; multiplying by 2 vertically stretches by 2; adding 3 shifts up by 3.

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